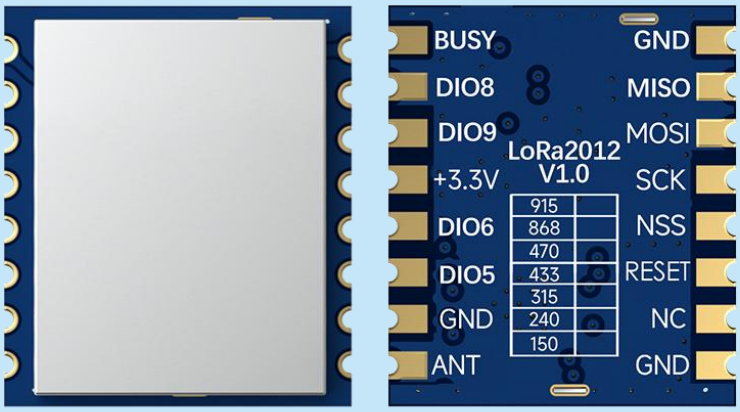
LoRa2012 Wireless Communication Module

- Sub-GHz band: 150-960MHz
- Up to 125Kbps@LoRa
- Multiple modulations: FSK, LoRa, OOK
- Compatible with multiple low-power wireless protocols: Amazon Sidewalk, Wireless M-BUS, Wi-SUN FSK
- Supports LoRaWAN protocol
- Supports RTToF ranging

Product Specification



Catalogue

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Note: Revision History

Revision	Date	Comment
V1.0	2026-6	First release

1. Descriptions

The LoRa2012 module is based on Semtech's LR2012 chip, which integrates the fourth-generation LoRa® IP core and supports sub-GHz and ISM frequency bands. It is fully backward compatible with previous LoRa devices, ensuring seamless interoperability with LoRaWAN® networks.

The LoRa2012 is manufactured and tested using a strictly lead-free process and complies with the RoHS and REACH environmental standards.

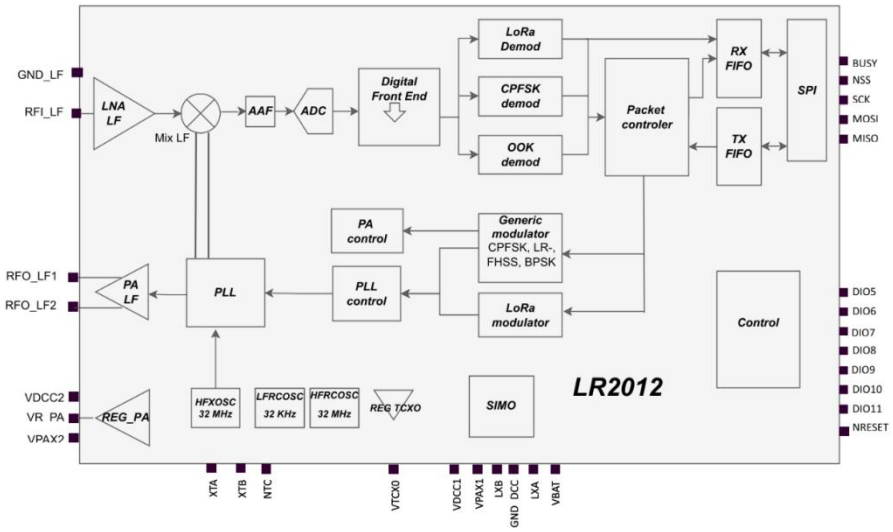
2. Features

- Sub-GHz Bands: 433/470/868/915MHz (Customizable range: 150~960 MHz)
- Transmission distance: >5000 meters at sub-GHz in open area
- Sub-GHz reception sensitivity: up to -143dBm@BW=62.5 KHz, SF=12
- Built-in electrostatic protection circuit
- LoRa: modulation rate up to 125 kbps
- 4th Generation LoRa IP technology
- Simultaneous reception with multiple Spreading Factors (SF)
- Enhanced CAD (Channel Activity Detection)
- Higher frequency offset tolerance (for harsh RF environments).
- Supports multiple communication protocols
- Transmit power is adjustable, up to 22dBm
- Sleep current ≤ 2μA
- Receive current < 7mA

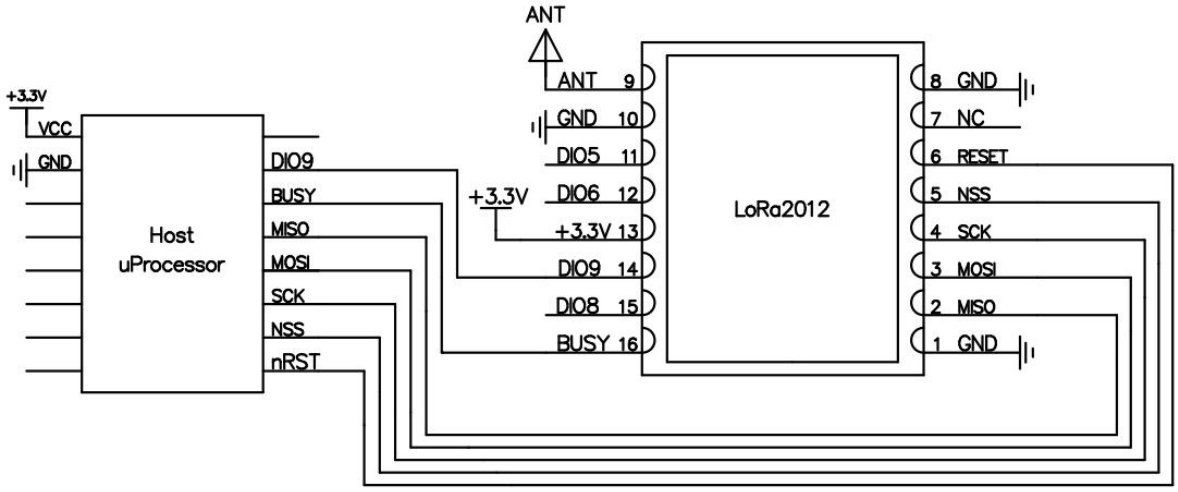
3. Applications

- Smart Home
- Smart agriculture
- Smart City
- Industrial Automation

4. Block Diagram



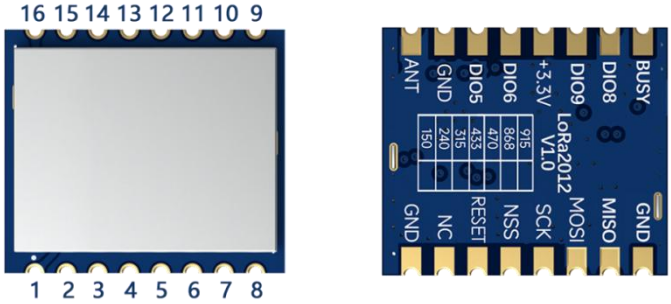
5. Typical application circuit



6. Electrical Characteristics

Parameters	Test condition	Min.	Typ.	Max	Unit
Voltage range		1.8	3.3	3.6	V
Operating Temperature		-40	25	85	℃
Maximum Input Signal			10		dBm
Current Consumption					
Transmit Current	@433MHz		< 110		mA
	@868/915MHz		< 120		mA
Receive Current	@3.3V,@DCDC,Sub-GHz		< 6		mA
	@3.3V,@LDO,Sub-GHz		< 9.3		mA
Sleep Current	@3.3V		≤2		uA
RF Parameters					
Frequency Range	@433MHz	400		460	MHz
	@470MHz	470		510	MHz
	@868MHz	850		890	MHz
	@915MHz	900		940	MHz
Transmit Power	@Sub-GHz	19	21	22	dBm
Receive Sensitivity	BW=62.5KHz, SF=12 @Sub-GHz		-143		dBm
	BW=125KHz, SF=10 @Sub-GHz		-136		dBm
Frequency Error	@Crystal		10		ppm
	@TCXO		0.5		ppm
Modulation Rate (@sub-GHz)	@LoRa	0.0458		125	Kbps
	@FSK,@433MHz-960MHz	0.476		101.5	Kbps

7. Pin definition



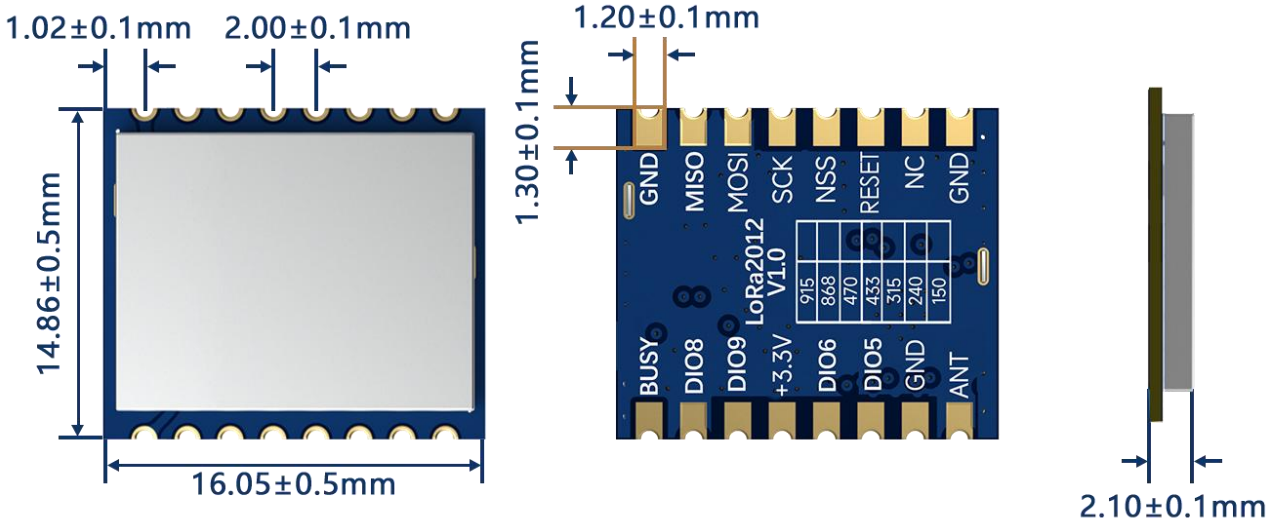
Pin NO.	Pin name	I/O	Description
1,8,10	GND		Connect to the negative power supply.
2	MISO	O	SPI data output
3	MOSI	I	SPI data input
4	SCK	I	SPI clock input
5	NSS	I	SPI chip select input
6	RESET	I	Reset trigger input, refer to the chip datasheet for details.
7	NC		
9	ANT		@sub-GHz band antenna interface, external 50-ohm antenna.
11	DIO5	IO	Multipurpose digital interface, refer to the chip datasheet for details.
12	DIO6	IO	Multipurpose digital interface, refer to the chip datasheet for details.
13	+3.3V	O	Connect to the positive power supply.
14	DIO9	IO	Multipurpose digital interface, refer to the chip datasheet for details.
15	DIO8	IO	Multipurpose digital interface, refer to the chip datasheet for details.
16	BUSY	O	Used for status indication, refer to the chip datasheet for details.

8. Power and Current Comparison Table

Power Levels of the DEMO Board		0	1	2	3	4	5	6	7	8	9
Sub-GHz Register Values		-11	-5	1	7	13	19	25	31	37	44
@433MHz @3.3V	Power(dBm)	-6	-3.4	0.6	2.4	5.3	8.2	11.8	15.0	18.0	21.4
	Current(mA)	10.0	10.8	11.8	13.8	15.7	22.4	23.5	30.0	50.3	104

Register Value 1: TxPower in the SetTxParams(0x0203) Command

9. Mechanism Dimension (Unit: mm)



Appendix 1: Function Demonstration Board



Note: Only one power supply method can be selected

1. Function Description

The LoRa2012 wireless module demo board primarily implements bidirectional communication (Master and Slave) in LoRa mode, along with sleep (Sleep), transmit power (TxTest), and receiver sensitivity (RxTest) testing functions at the modulations of LoRa.

Note: Transmit power and receiver sensitivity must be tested with instruments.

2. Key Functions

The demo board has three buttons: SET, UP, and DOWN. Their functions are as follows:

Buttons	Functions
SET Key Short Press	Confirm or enter the next level of the interface
SET Key Long Press	Return to the previous interface
UP Key Short Press	Move cursor up or increase parameter by 1
UP Key Long Press	Increase parameter
DOWN Key Short Press	Move cursor down or decrease parameter by 1
DOWN Key Long Press	Decrease parameter

Operation Method:

With normal power supply, toggle the power switch to turn on the device. The LCD screen on the demo board will display the "Setting" interface. Briefly press the UP or DOWN key to move the cursor. When the cursor is on the parameter you wish to modify, press the SET key briefly to make the cursor flash. Then, use the UP or DOWN key to adjust the parameter. Press the SET key again to confirm the modification.

To enter the test function corresponding to the "Mode" parameter, long-press the SET key. Briefly press the SET key to return to the "Setting" interface.

3. Setting Interface Description

Parameter Name	Description	LoRa Mode
Mod	Modulation	LoRa
Mode	Function	Master, Slave, TXTEST, RXTEST, SLEEP
Band	Frequency Band	LF: 150-960MHz
FREQ	Frequency(MHz)	Set value(MHz)
POWER	TX Power	0-9 Levels
SF(LoRa)	Spreading Factor	5-12
BW(LoRa)	Bandwidth	62.5/125/250/500/1000(Sub-GHz)
CR(LoRa)	Coding Rate	4/5-4/8
DR_BW(FLRC)	Data rate, Bandwidth(Mbps)	N/A
CR(FLRC)	Coding Rate	N/A
Shape(FLRC)	Pulse shape	N/A

4. Communication Test

Prepare two LoRa2012 demo boards, and select the modulation: LoRa . Set one to Master mode as the communication host, and the other to Slave mode as the communication slave. (Set the LoRa communication parameters: FREQ, SF, BW, and CR to be the same; otherwise, communication will fail.)

On the communication interface screen, the Tx_cnt and Rx_cnt values will keep increasing. (The red indicator light represents a successful transmission, and the blue indicator light represents a successful reception.)


Slave
Tx_Cnt:2
Rx_Cnt:3
SET
UP
DOWN
Master
Tx_Cnt:3
Rx_Cnt:2
SET
UP
DOWN

Appendix 2: SMD Reflow Chart

Below reflow profile is recommended for SMT technology:

